

Maxime Newman Nereyabagabo

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EDUCATION

Simon Fraser University

Jan. 2024 – Aug. 2028

Bachelor of Science, Computing Science

Burnaby, BC

- Coursework: Machine Learning, Data Structures, Algorithms, Databases, Statistics, Networking, Systems Programming

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C++, C#, HTML/CSS, Bash, SQL

ML & AI: Scikit-learn, LangGraph, Pydantic, Pandas, NumPy, LangSmith

Frameworks & Libraries: Next.js, FastAPI, Django

Databases & Caching: PostgreSQL, Redis, Supabase

Cloud, DevOps & Automation: Docker, GitHub Actions, AWS, GCP, Azure, Git, CI/CD

Testing: Pytest, JUnit, Mockito, GoogleTest

RELEVANT EXPERIENCE

SFU Cybersecurity

May. 2026 – Present

Research Assistant

Burnaby, BC

- Co-authoring research paper on automated multi-agent security vulnerability remediation, contributing pipeline architecture and end-to-end evaluation framework across **120** Java CVE benchmarks
- Designed and evaluated a **4-stage** agentic security pipeline (Finder → Exploiter → Patcher → Verifier) integrating **IRIS/CodeQL** static analysis with **LLM-driven** exploit and patch generation, benchmarked end-to-end on **120** Java CVE projects
- Reduced pipeline iteration latency by **72%** and cut redundant static-analysis recomputation by **85%** by implementing a state-based SARIF caching layer for the Finder stage of a **LangGraph**-based agentic pipeline
- Increased pipeline coverage from **73%** to **100%** across a **120-project** Java CVE benchmark by introducing graceful fallback handling for missing cache artifacts, eliminating startup failures on previously-unseen projects
- Reduced redundant analysis on retry paths by **40%** by decoupling source-tree extraction from cache invalidation in the Finder dispatch logic, collapsing 4 incoherent flag-state combinations into 2 well-defined modes
- Cut per-experiment setup time by **85%** by refactoring CLI flag semantics, eliminating manual code edits previously required to toggle between cached and fresh pipeline modes

SKC Engineering Ltd

Jan. 2026 – May 2026

Data Engineer

Surrey, BC

- Classified **15,639 SharePoint** files into **6 welding-process** categories with **70% requiring no manual review** by building an authenticated PowerShell pipeline via Azure App Registration, delivering a cleaned dataset to the ML team for fine-tuning
- Saved an estimated **\$15,000+** in engineer time by building structured CSV logging with company-level and file-level outcomes, enabling full auditability and data quality tracking across parallel letter-batch migration runs
- Migrated **29,000+ rows** across **87 tables** from legacy Linode MySQL to managed DigitalOcean MySQL with **100% parity** by engineering a topological-sort foreign-key resolver, idempotent batched inserts, and symmetric pre/post-migration primary-key validation

SKC Engineering Ltd

Sept. 2025 – Jan. 2026

AI Engineer

Surrey, BC

- Built a full-stack AI-powered cost estimation platform with **GPT-4o mini** and **LangGraph**, using intent-based routing, deterministic tool execution, and structured state management to deliver consistent, auditable outputs
- Cut LLM inference cost per request by **20%** by stabilizing input size across long parameter-gathering sessions through rolling context windows, automatic summarization, and state-first prompting
- Improved model consistency across deployments with **50+ LangSmith eval cases**, using human-in-the-loop interrupts and a regression monitoring pipeline to detect behavioral drift
- Achieved **sub-300 ms API response times** for core authenticated endpoints by deploying to Fly.io using Docker multi-stage builds, GitHub Actions CI/CD, JWT-based access control, and idempotent APIs supporting a persistent multi-user workspace

SFU CS ARCH Group

Jan 2025 – Present

Research Assistant

Burnaby, BC

- Researched attention mechanisms and transformer architectures, focusing on how large-scale matrix multiplications dominate training and inference of LLMs.
- Authored 'read-along guide' on GPU-accelerated matrix multiplication kernel optimizations in CUDA, outlining concepts based on GPU multi-threading and execution models (thread blocks, warps, SM scheduling).
- Leveraged CUDA programming to design kernels in Linux environments to achieve 93.7% of cuBLAS using NVIDIA Nsight Compute (NCU) for analysis. Inspected PTX assembly output to further unravel optimizations, understanding IR passes done by the compiler itself.

Zebra Robotics Surrey

Jan. 2025 – Jun. 2025

Robotics Instructor

Surrey, BC

- Taught programming and robotics fundamentals using Python, Arduino, and microcontroller labs, enabling students to successfully program motors and sensors and complete functional robot prototypes
- Led student robotics projects from concept design through testing, guiding teams in mechanical iteration and debugging with programmable hardware, resulting in completed robots that met project goals and were showcased in a school exhibition

PROJECTS

FraudLens | *Python, Pandas, Scikit-learn, NumPy*

[GitHub](#) | SFU DSSS ML Hackathon

- Won **1st place** at the SFU DSSS ML Hackathon by building a multi-class fraud detection pipeline using Python, Pandas, and scikit-learn under tight time constraints
- Engineered 5+ predictive features using Pandas and NumPy, including **balance mismatch flags and error signals** to expose accounting inconsistencies, improving recall on rare minority fraud classes
- Trained a class-weighted **Random Forest** tuned against **Macro F1**, with a constraint-based data processing pipeline enforcing consistent types and flagging transaction impossibilities

Goblin's Keep | *Java, Maven, JUnit, Mockito*

[GitHub](#) | [Play Game](#)

- Led backend development of a 2D tile-based escape game, architecting game logic and AI systems, establishing CI/CD pipelines with GitHub Actions, and coordinating code reviews across a team of developers
- Implemented dynamic goblin AI using a modified **A* pathfinding algorithm** with randomized patrol logic, enhancing difficulty balance and replayability
- Achieved **96% line coverage** and **92% branch coverage** with a **JUnit/Mockito test suite** ensuring correctness across all gameplay interactions

OTHER EXPERIENCES

Simon Fraser University

Jan. 2025 – Apr. 2025

Calculus Teaching Assistant

Burnaby, BC

- Led weekly peer-led sessions for a dedicated group of students, fostering collaborative learning and long-term academic growth
- Explained key concepts through interactive discussions and guided problem-solving sessions
- Reinforced personal mastery of calculus through teaching, feedback, and real-time troubleshooting

Simon Fraser University

Aug. 2024 – Sept. 2024

Hive Leader

Burnaby, BC

- Organized and led events to welcome and engage new students
- Collaborated with other Hive Leaders to plan and execute group activities
- Supported and mentored HIVE volunteers aspiring to become Hive Leaders
- Led orientation sessions, ensuring new students had a smooth transition

SFU Recreation

Apr. 2024 – Aug. 2024

Recreation Assistant

Burnaby, BC

- Front Desk Reception: Managed check-ins, answered inquiries, and kept daily operations running smoothly
- Customer Service: Built relationships with clients, addressed concerns, and made sure every member felt welcomed
- Fitness Class Assistant: Supported instructors with class setup and equipment while keeping participants motivated and sessions running safely
- Intramural Soccer League Assistant: Helped organize and run a soccer league, handling scheduling, officiating, and ensuring games ran smoothly